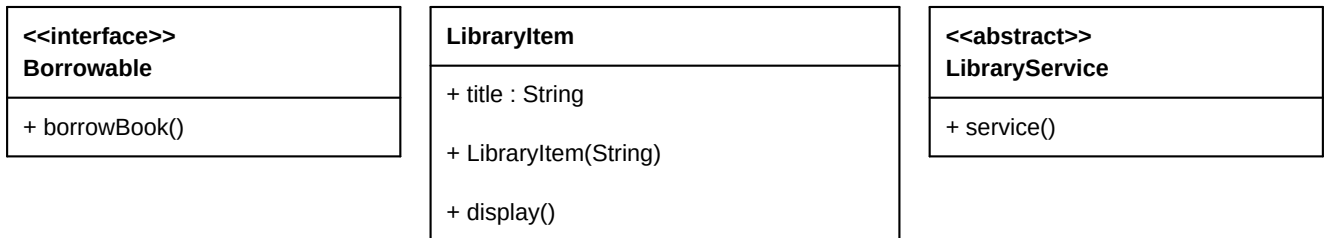


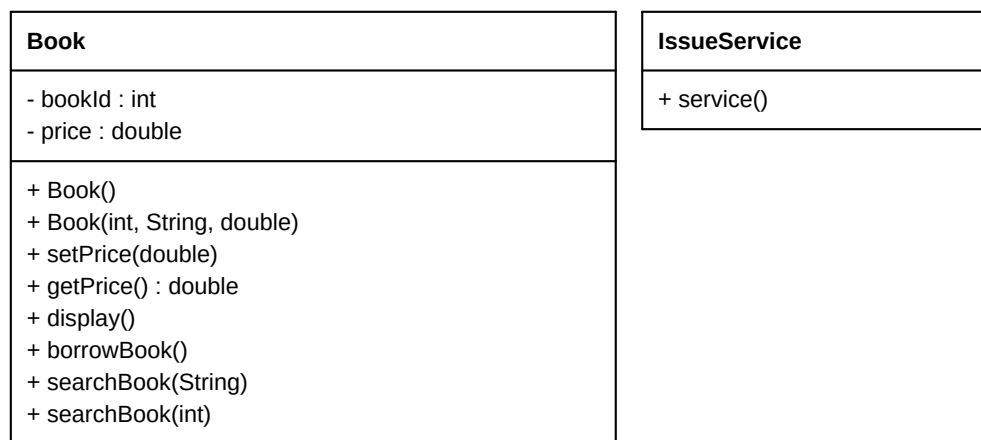
LIBRARY MANAGEMENT SYSTEM

Java OOP Program — Classes, Objects, Constructors, Encapsulation, Inheritance, Polymorphism, Abstraction and Interface

UML CLASS DIAGRAM



implements extends extends



Source Code

```
// Interface
interface Borrowable {
    void borrowBook();
}

// Base class
class LibraryItem {
    String title;

    LibraryItem(String title) {
        this.title = title;
    }

    void display() {
        System.out.println("Library Item: " + title);
    }
}

// Inheritance + Method Overriding
class Book extends LibraryItem implements Borrowable {
    private int bookId;
    private double price;

    // Default constructor
    Book() {
        super("Unknown");
        bookId = 0;
        price = 0;
    }

    // Parameterized constructor
    Book(int bookId, String title, double price) {
        super(title);
        this.bookId = bookId;
        this.price = price;
    }
}
```

```

    }

    // Encapsulation
    void setPrice(double price) {
        this.price = price;
    }

    double getPrice() {
        return price;
    }

    @Override
    void display() {
        System.out.println("Book ID : " + bookId);
        System.out.println("Title : " + title);
        System.out.println("Price : " + price);
    }

    @Override
    public void borrowBook() {
        System.out.println("Book borrowed successfully.");
    }

    // Method Overloading
    void searchBook(String title) {
        System.out.println("Searching book: " + title);
    }

    void searchBook(int bookId) {
        System.out.println("Searching book ID: " + bookId);
    }
}

// Abstraction
abstract class LibraryService {
    abstract void service();
}

class IssueService extends LibraryService {
    void service() {
        System.out.println("Book issue service");
    }
}

public class LibraryManagementSystem {
    public static void main(String[] args) {

        // Class and Object
        Book book = new Book(101, "Java Programming", 450);

        book.display();

        // Encapsulation
        book.setPrice(500);
        System.out.println("Updated Price: " + book.getPrice());

        // Method Overloading
        book.searchBook("Java Programming");
        book.searchBook(101);

        // Interface
        book.borrowBook();

        // Abstraction
        LibraryService service = new IssueService();
        service.service();
    }
}

```

Compilation and Execution Commands

Save as: LibraryManagementSystem.java
Compile: javac LibraryManagementSystem.java
Execute: java LibraryManagementSystem

Expected Output — Library Book and OOP Operations

Book ID : 101
Title : Java Programming
Price : 450.0
Updated Price: 500.0
Searching book: Java Programming
Searching book ID: 101
Book borrowed successfully.
Book issue service

Concepts Demonstrated

Concept	Where it is used
Class & Object	LibraryItem, Book and the object book
Constructor	Book() and Book(int, String, double)
Encapsulation	private price with setPrice() and getPrice()
Inheritance	Book extends LibraryItem
Method Overriding	Book overrides display()
Method Overloading	searchBook(String) and searchBook(int)
Interface	Book implements Borrowable
Abstraction	LibraryService and IssueService

Result

Thus, the Library Management System was implemented successfully to demonstrate the basic Object-Oriented Programming concepts of Java using classes, objects, constructors, encapsulation, inheritance, polymorphism through overloading/overriding, abstraction, and interfaces.

Important Viva Questions

1. What is inheritance?

Inheritance allows a child class to acquire properties and methods from a parent class.

2. What is an interface?

An interface defines methods that an implementing class must provide.

3. What is encapsulation?

Encapsulation keeps data private and accesses it through methods such as setters and getters.

4. What is method overloading?

Using the same method name with different parameter lists.

5. What is abstraction?

Showing essential functionality while hiding implementation details.